

Report 1: Thematic Analysis — Research Motivation Categories

Analyst: AI (MiniMax M2.7, via Zo Computer)

Source: 27 raw OCR text files from Documents/OCR 1.1/Raw/

Date: 2026-04-28

Section A: Category Counts and Poster Lists

Category | Count | Posters |

|—|—|—|

Basic Biology | 11 | Brown et al. (2025) - CCC SP25, Ford et al. (2023) - CCC SP23, Godoy-Pena et al. (2025) - CCC FA25, Henriquez et al. (2025) - COD WI24, Holmes (2025) - CCC SP25, Logan et al. (2025) - CCC FA25, Luera et al. (2025) - CCC FA25, Meraz et al. (2023) - CCC SP23, Sakana et al. (2025) - CCC SP25, Tuttle et al. (2025) - CCC SP25, Zlaket et al. (2023) - CCC SP23 |

Neurological & Psychiatric Disease | 4 | Avetisyan et al. (2023) - CCC SP23, Cabral et al. (2025) - CCC FA25, Gill & Alcazar (2025) - CCC FA25, Haubelt & Alcazar et al. (2025) - CCC FA25 |

Cancer | 1 | Paramo-Ojeda et al. (2025) - CCC SP25 |

Metabolic & Digestive Disease | 9 | Berbouti et al. (2025) - CCC FA25, Camara et al. (2025) - LU SP25, Dehuelbes et al. (2025) - LU SP5, Diaz et al. (2025) - CCC SP25, Lemus et al. (2025) - FA25 CCC, Nii et al. (2025) - CCC FA25, Ojala et al. (2025) - LU SP25, Pedireddi et al. (2025) - CCC SP25, Rodriguez (2025) - CCC SP25 |

Inherited / Genetic Disorder | 2 | Paderna et al. (2025) - CCC SP25, Trevino et al. (2022) - CCC SP22 |

Other | 0 | |

Total posters analyzed: 27

Section B: Category Definitions

Basic Biology

Research focused on understanding fundamental biological mechanisms — such as gene expression patterns, regional specialization of tissues, signaling pathways, or developmental processes — without explicitly targeting a specific human disease. These posters use model organisms to explore how genes and cells normally function. Example topics: regional gut gene expression, immune pathway profiling, neuropeptide signaling biology, or paralog/ortholog characterization.

Neurological & Psychiatric Disease

Research motivated by understanding the molecular basis of brain or nervous system disorders. These posters investigate genes and pathways linked to cognitive, behavioral, or neuropsychiatric conditions in humans. Example topics: autism spectrum disorder, Parkinson’s disease, serotonin receptor signaling, dopamine pathways, tryptophan-kynurenine metabolism, or Rett syndrome.

Cancer

Research motivated by understanding cancer biology, specifically oncogenes, tumor suppressor genes, cell cycle regulation, or cancer-related pathways. These posters frame the work around uncontrolled cell growth, tumor development, or cancer risk. Example topics: KRAS oncogene, mismatch repair genes, tumor suppressors (APC, BRCA2), or colorectal cancer models.

Metabolic & Digestive Disease

Research motivated by understanding metabolic or digestive disorders, including conditions affecting the liver, gut, or nutrient processing. These posters focus on pathways such as carbohydrate metabolism, fatty acid oxidation, acid reflux, or nutrient absorption. Example topics: acid reflux disease, fatty acid metabolism disorders, galactosemia, amylase gene expression, or trehalose catabolism.

Inherited / Genetic Disorder

Research motivated by understanding specific inherited or rare genetic disorders. These posters explicitly name and characterize a clinical genetic condition and use the model organism to understand its mechanisms. Example topics: Zellweger Spectrum Disorder, oculocutaneous albinism (OCA4), MCADD, or other rare inborn errors of metabolism.

Other

Research that does not fit into any of the five primary categories. This may include posters focused purely on technical method validation, broadly comparative biology without disease framing, or topics that cross multiple categories without a dominant motivation.

Section C: Per-Poster Evidence for Category Assignment

For each poster, a bullet-point list provides specific text evidence supporting the category assignment. For posters with elements of multiple categories, the tie-breaking rationale is noted.

Avetisyan et al. (2023) - CCC SP23

Assigned Category: Neurological & Psychiatric Disease

Note: This poster also showed elements of: Basic Biology (scores: {'Basic Biology': 2}). Category assigned based on dominant motivation.

- **“Expression Analysis of Autism Related Genes Across the D. Melanogaster Midgut”*
- **“gut-microbiome-brain axis, have been identified as autism related.”*

Berbouti et al. (2025) - CCC FA25

Assigned Category: Metabolic & Digestive Disease

Note: This poster also showed elements of: Basic Biology (scores: {'Basic Biology': 3}). Category assigned based on dominant motivation.

- **“Trehalose is a critical sugar in Drosophila melanogaster, in which its breakdown by”*
- **“study provides insight into the regulation of carbohydrate metabolism in flies and”*
- **“trehalase provides glucose for energy metabolism. The pathway of trehalose”*
- **“posterior midgut, TRET1 was enriched in the copper cell region, and GLUT1 was”*

Brown et al. (2025) - CCC SP25

Assigned Category: Basic Biology

Note: This poster also showed elements of: Neurological & Psychiatric Disease, Metabolic & Digestive Disease (scores: {'Neurological & Psychiatric Disease': 1, 'Metabolic & Digestive Disease': 5}). Category assigned based on dominant motivation.

- **“ortholog being ATP6VOC, which encodes a subunit of the vacuolar H'-ATPase (V-ATPase), necessary for”*
- **“hormonal signaling pathways, vhal6-1 emerges as a core integrator of digestive processes and systemic”*

- *“(V-ATPase, which is a channel protein that encourage multiple functions. Within this we identified paralogs to”
- *“(processes are closely regulated by gut-brain sensory signaling, which adapts feeding behavior throughout the”
- *“(melanogaster midgut, it focus on acidification processes that influence metabolic gene regulation. These”
- *“(exponentially expressed in the copper region with a”

Cabral et al. (2025) - CCC FA25

Assigned Category: Neurological & Psychiatric Disease

Note: This poster also showed elements of: Inherited / Genetic Disorder, Basic Biology (scores: {‘Inherited / Genetic Disorder’: 1, ‘Basic Biology’: 2}). Category assigned based on dominant motivation.

- *“(The methyl-CpG binding protein 2 (MECP2) gene is essential for neuronal”
- *“(are linked to Rett syndrome and related neurodevelopmental disorders; however, the”
- *“(beta 3 (Gabbrb3) and brain derived neurotrophic factor (Bdnf). Mutstions in MECP2”
- *“(beta 3 (Gabbrb3) and brain derived neurotrophic factor (Bdnf). Mutstions in MECP2”
- *“(in the same regions, while Bdnf shows higher expression in the prefrontal cortex”
- *“(including the hippocampus, cerebellum, cortex, striatum, and thalamus. Expression”

Camara et al. (2025) - LU SP25

Assigned Category: Metabolic & Digestive Disease

Note: This poster also showed elements of: Neurological & Psychiatric Disease, Basic Biology (scores: {‘Neurological & Psychiatric Disease’: 1, ‘Basic Biology’: 3}). Category assigned based on dominant motivation.

- *“(— How is starch digestion regulated in the human gut—beyond just gene copy”
- *“(We studied expression of three fly amylase genes: Amy-d, Amy-p,”
- *“(like diabetes and Alzheimer’s (Leue, 2016).”
- *“(Regions compared: anterior (al) vs. copper cell (Cu).”

Dehuelbes et al. (2025) - LU SP5

Assigned Category: Metabolic & Digestive Disease

Note: This poster also showed elements of: Inherited / Genetic Disorder, Basic Biology (scores: {‘Inherited / Genetic Disorder’: 1, ‘Basic Biology’: 2}). Category assigned based on dominant motivation.

- *“(Precursor for synthesis of fatty acids, cholesterol, and”
- *“(target for treating metabolic disorders in humans.”

Diaz et al. (2025) - CCC SP25

Assigned Category: Metabolic & Digestive Disease

Note: This poster also showed elements of: Neurological & Psychiatric Disease, Basic Biology (scores: {‘Neurological & Psychiatric Disease’: 1, ‘Basic Biology’: 2}). Category assigned based on dominant motivation.

- *“(the Copper Region of the Drosophila Viidgut »”
- *“(crucial role of acidification within copper cells; therefore,”

Ford et al. (2023) - CCC SP23

Assigned Category: Basic Biology

- *Some Cytochrome P450 genes play an important role in the development”
- *found in al, p1 and p2 4 midgut regions of Drosophila”

Gill & Alcazar (2025) - CCC FA25

Assigned Category: Neurological & Psychiatric Disease

Note: This poster also showed elements of: Basic Biology (scores: {‘Basic Biology’: 1}). Category assigned based on dominant motivation.

- *Differential Expression of Serotonin Receptor Genes Htr1a and Htr2a”
- *5-HTR1A receptors which are encoded by HT1A are expressed more in cortical”
- *Differential Expression of Serotonin Receptor Genes Htr1a and Htr2a”
- *ies the prefrontal cortex and the striatum In the brain the role of the”
- *understand mental health disorders (1). Htr2a on the otherhand isa”
- *in the Prefrontal Cortex and Striatum”
- *in the Prefrontal Cortex and Striatum”

Godoy-Pena et al. (2025) - CCC FA25

Assigned Category: Basic Biology

Note: This poster also showed elements of: Neurological & Psychiatric Disease, Metabolic & Digestive Disease (scores: {‘Neurological & Psychiatric Disease’: 2, ‘Metabolic & Digestive Disease’: 1}). Category assigned based on dominant motivation.

- *This Neuropeptide in flies”
- *gut-brain region. Finding orthologs to further the evidence of”
- *influence the drosophila fly’s circadian rhythm. The genes”
- *Regulation of Sleep Quality: Analysis of Gene Expression Using”
- *gut-brain region. Finding orthologs to further the evidence of”
- *copper region of the drosophila fly midgut, which is responsible”
- *- Confirmed CChA2-R presence in the copper midgut region”

Haubelt & Alcazar et al. (2025) - CCC FA25

Assigned Category: Neurological & Psychiatric Disease

Note: This poster also showed elements of: Basic Biology (scores: {‘Basic Biology’: 1}). Category assigned based on dominant motivation.

- *to the neuropsychiatric disorder Parkinson’s Disease and wanted”
- *Disease, Schizophrenia, and Post-traumatic Stress Disorder are”
- *Exploring DRD2: The Association of PD Risk with Dopamine”
- *expressed in the striatum part of the mice brain. The striatum is a”
- *to the neuropsychiatric disorder Parkinson’s Disease and wanted”
- *expressed in the striatum part of the mice brain. The striatum is a”

Henriquez et al. (2025) - COD WI24

Assigned Category: Basic Biology

Note: This poster also showed elements of: Metabolic & Digestive Disease (scores: {‘Metabolic & Digestive Disease’: 1}). Category assigned based on dominant motivation.

- *6 Immune Pathways Exhibit Similar Patterns of Gene Expressions”
- *The different cell types that reside within the midgut may play”
- *immune pathways to regulate the gut microbiome.”
- *to the copper region, unlike the rest of the pathways. Due to”

- *“midgut regions of *Drosophila melanogaster* characterized by”

Holmes (2025) - CCC SP25

Assigned Category: Basic Biology

Note: This poster also showed elements of: Neurological & Psychiatric Disease, Metabolic & Digestive Disease (scores: {‘Neurological & Psychiatric Disease’: 2, ‘Metabolic & Digestive Disease’: 1}). Category assigned based on dominant motivation.

- *“was significantly expressed in the Fe region. A paralog and ortholog gene”
- *“differential expression between the regions. Through that we will do further”
- *“was significantly expressed in the Fe region. A paralog and ortholog gene”

Lemus et al. (2025) - FA25 CCC

Assigned Category: Metabolic & Digestive Disease

Note: This poster also showed elements of: Cancer, Basic Biology (scores: {‘Cancer’: 1, ‘Basic Biology’: 4}). Category assigned based on dominant motivation.

- *“the regional expression patterns of two genes both involved in starch and”
- *“the amylase family would share a similar function given their lineage. Upon”
- *“digestion as well as metabolic disorders. Our researched aimed to compare”
- *“starch hydrolysis, the breakdown of starch into glucose and maltose through the addition of”
- *“with the hydrolysis of dietary starch, is significantly differentially expressed in”

Logan et al. (2025) - CCC FA25

Assigned Category: Basic Biology

Note: This poster also showed elements of: Cancer (scores: {‘Cancer’: 1}). Category assigned based on dominant motivation.

- *“fly ortholog of the human ACE gene. We analyzed four ANCE paralogs”
- *“- Utilized SciServer, under Differential Expression with DESeq2, to find if the”
- *“fly ortholog of the human ACE gene. We analyzed four ANCE paralogs”
- *“- To gather more supporting information, we looked at the different cell types”
- *“development. Since the ACE gene 1s a regulator of fluidity and”
- *“expression patterns. Across different midgut regions, we observed distinct”

Luera et al. (2025) - CCC FA25

Assigned Category: Basic Biology

- *“We analyzed Neuropeptide F (NPF) and its receptor NPFR mm *Drosophila*”
- *“Gene Identification: Using FlyBase, orthologs of mammalian NPY and its”
- *“Differential expression analysis revealed that NPF and NPFR were among the most significantly regulated genes across mudzut regions (adjusted p-value < 0.03,”
- *“differentially expressed genes between mudzut regions or cell types. Genes were”
- *“learning/memory, and circadian regulation.”
- *“and sleep behaviors m *Drosophila melanogaster* and 1s analogous to mammalian”
- *“may serve as fimctional hubs for feeding regulation. Unexpectedly, some midgut regions”

Meraz et al. (2023) - CCC SP23

Assigned Category: Basic Biology

Note: This poster also showed elements of: Cancer, Metabolic & Digestive Disease (scores: {‘Cancer’: 1, ‘Metabolic & Digestive Disease’: 1}). Category assigned based on dominant motivation.

- *“which is the human ortholog of Src42A, is involved in”
- *“differential expression of the SRC gene in the Cu and al”
- *“disease-related genes in drosophila contain paralogs in”
- *“related to cell growth and development. The FRK gene,”
- *“Tyrosine Kinase Expression in the Midgut Suggests a Role in the Copper Region”

Nii et al. (2025) - CCC FA25

Assigned Category: Metabolic & Digestive Disease

Note: This poster also showed elements of: Basic Biology (scores: {‘Basic Biology’: 1}). Category assigned based on dominant motivation.

- *“entry point linking carbohydrate metabolism to lipid synthesis. While”
- *“converts acetyl-Cod to malonyl-Cod, committing cells to fatty acid synthess.”

Otala et al. (2025) - LU SP25

Assigned Category: Metabolic & Digestive Disease

Note: This poster also showed elements of: Cancer, Inherited / Genetic Disorder, Basic Biology (scores: {‘Cancer’: 1, ‘Inherited / Genetic Disorder’: 4, ‘Basic Biology’: 3}). Category assigned based on dominant motivation.

- *“Fly Gut is a good model for fatty acid metabolism disorders”
- *“reflects the posterior midguts role in nutrient absorption and energy production.”

Paderna et al. (2025) - CCC SP25

Assigned Category: Inherited / Genetic Disorder

Note: This poster also showed elements of: Metabolic & Digestive Disease, Basic Biology (scores: {‘Metabolic & Digestive Disease’: 1, ‘Basic Biology’: 2}). Category assigned based on dominant motivation.

- *“with oculocutaneous albinism type 4, a”
- *“(OCA4) manifests as reduced pigment in the skin,”
- *“Albinism and Melanogenesis Pathway:”
- *“mutations in the gene SLC45A2, a proton”

Paramo-Ojeda et al. (2025) - CCC SP25

Assigned Category: Cancer

Note: This poster also showed elements of: Metabolic & Digestive Disease, Inherited / Genetic Disorder, Basic Biology (scores: {‘Metabolic & Digestive Disease’: 1, ‘Inherited / Genetic Disorder’: 1, ‘Basic Biology’: 4}). Category assigned based on dominant motivation.

- *“~~ Model for KRAS-Driven Colon Cancer”
- *“lead to tumor development and are often associated with poor”
- *“most frequently mutated oncogenes in human cancers is”
- *“~~ Model for KRAS-Driven Colon Cancer”
- *“Ras85D Expression in Drosophila Midgut Highlights a”
- *“~~ Model for KRAS-Driven Colon Cancer”
- *“including BRCA2, APC, Pi3K92E (PIK3CA), MLH1, and PMS2,”
- *“including BRCA2, APC, Pi3K92E (PIK3CA), MLH1, and PMS2,”

Pedireddi et al. (2025) - CCC SP25

Assigned Category: Metabolic & Digestive Disease

Note: This poster also showed elements of: Basic Biology (scores: {'Basic Biology': 2}). Category assigned based on dominant motivation.

- *"15 genes In the starch and sucrose metabolic pathway. We found 7 genes"
- *"Beyond the Anterior: Amylase Gene Expression Suggests Widespread"
- *"Polysaccharide Digestion in the Drosophila Midgut |"
- *"the P region was also responsible for nutrient absorption (Marianes,"

Rodriguez (2025) - CCC SP25

Assigned Category: Metabolic & Digestive Disease

Note: This poster also showed elements of: Inherited / Genetic Disorder, Basic Biology (scores: {'Inherited / Genetic Disorder': 2, 'Basic Biology': 1}). Category assigned based on dominant motivation.

- *"Galactosemia is a rare metabolic disorder that results from deficiencies in the"
- *"the disease galactosemia, which prevents the body from properly metabolising"

Sakana et al. (2025) - CCC SP25

Assigned Category: Basic Biology

- *"melanogaster known for its role in tissue patterning, stem cell"
- *"Differential Expression of the TALE Homeobox Genes in"
- *"Integration Site genes (MEIS1 and MEIS2), orthologs of"
- *"Differential Expression of the TALE Homeobox Genes in"
- *"differentiation are often orchestrated by transcription factors."
- *"gene paralogs in Drosophila. Drosophila melanogaster serves as"
- *"Gene expression patterns that control development and"
- *"Extension (TALE) homeobox genes across the midgut regions."

Trevino et al. (2022) - CCC SP22

Assigned Category: Inherited / Genetic Disorder

Note: This poster also showed elements of: Metabolic & Digestive Disease, Basic Biology (scores: {'Metabolic & Digestive Disease': 6, 'Basic Biology': 5}). Category assigned based on dominant motivation.

- *"Drosophila Melanogaster a Good Model System of Zellweger Spectrum Disorder"
- *"ABCD1, ACBDS, ACOX1, DNM1L, HSD17B4, and SCP2 are listed as peroxisomal disorders"
- *"ZSD is a rare inherited disorder characterized by the absence/reduction of functional"
- *"Zellweger Spectrum Disorder (ZSD) is a genetic disorder that is caused by the mutation"
- *"Zellweger Spectrum Disorder (ZSD) is a genetic disorder that is caused by the mutation"

Tuttle et al. (2025) - CCC SP25

Assigned Category: Basic Biology

Note: This poster also showed elements of: Neurological & Psychiatric Disease, Metabolic & Digestive Disease, Inherited / Genetic Disorder (scores: {'Neurological & Psychiatric Disease': 3, 'Metabolic & Digestive Disease': 1, 'Inherited / Genetic Disorder': 1}). Category assigned based on dominant motivation.

- *"This study investigates the differential expression of"

Zlaket et al. (2023) - CCC SP23

Assigned Category: Basic Biology

Note: This poster also showed elements of: Neurological & Psychiatric Disease, Metabolic & Digestive Disease, Inherited / Genetic Disorder (scores: {'Neurological & Psychiatric Disease': 1, 'Metabolic & Digestive Disease': 4, 'Inherited / Genetic Disorder': 4}). Category assigned based on dominant motivation.

- **“their function in flies and human orthologs, we were able to identify”*
- **“contrasted by the differential expression graphs in Table 1 as they”*
- **“ScpX: A Peroxisomal Gene and its Distinctive Paralogs”*
- **“to examine mutational effects and gene expression in many developmental”*
- **“midgut regions, particularly the”*